

CANADIAN AIRLINE LOYALTY PROGRAM

Churn Prediction & Retention Strategy

Business Analytics Competition — Technical Report | June 2026 | Dataset: ~16,700 Canadian loyalty members, 2012–2018

5.2× Precision lift over random PR AUC	50% Churners caught in top 10% Recall	30:1 Return on VIP retention spend ROI estimate
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Executive Summary

The airline's loyalty program retains 13,191 active members, but the marketing team had no reliable way to identify who was drifting before they cancelled. Cancellation is a lagging signal — by the time it appears in the data, the retention window has closed.

This report describes a machine learning system that predicts churn six months before cancellation occurs, using 41 behavioral and demographic features. The output is a risk score for every active member, mapped to one of six actionable segments. Three findings stand out:

- **CLV does not predict churn.** High-value members disengage at the same rate as low-value members when their recent behavior deteriorates. The model catches what CLV misses.
- **Consecutive zero-flight months is the single strongest churn signal**, more predictive than total spend, tenure, or demographics combined.
- **The top 10% riskiest members capture 50% of all churners.** That 5× lift concentrates retention spend where it matters most.

Three actions follow directly from the analysis:

- **Protect 346 VIP members immediately.** A \$52K personal outreach effort protects an estimated \$1.6M in CLV — a 30:1 return if 30% are retained.
- **Automate re-engagement for 3,100 cooling members.** Lounge passes and points reminders, fully automated, for less than \$10K annually.
- **Stop discounting healthy high-value members.** 2,711 members with 0% predicted churn risk still receive promotional emails that erode margin with no retention benefit.

1. Problem Framing

The loyalty program tracked one retention metric: Customer Lifetime Value (CLV). CLV measures past profitability but tells a manager nothing about who is cooling off right now. A member with \$20,000 in historical spend who last flew eight months ago looks identical to an actively flying high-value member — until it's too late to intervene.

Outreach decisions were made manually, reactively, and — by definition — too late. By the time cancellation appeared in the records, the member had already made their decision. A retention call at that point cannot change it.

2. Data & Cleaning

The dataset contains four source files covering approximately 16,700 Canadian airline loyalty members from 2012 to 2018: Customer Loyalty History (demographics, CLV, card tier, cancellation dates), Customer Flight Activity (monthly flights, distance, points earned and redeemed), a Calendar dimension table, and a data dictionary.

Data cleaning required five substantive decisions — each made deliberately, not mechanically:

- **3,871 duplicate rows found.** 1,922 were exact copies and were dropped. The remaining 1,949 were partial-month entries where a member's activity was split across two rows. These summed correctly, so we aggregated by member/year/month rather than dropping them. Dropping would have understated total activity.
- **20 negative salary values** — clearly data entry errors. Corrected by taking absolute value rather than removing the records.
- **Country column dropped** — every single row was Canada, zero variance, zero signal.
- **Null cancellation date = active member.** Not a bug. A null means the member has not cancelled. Flagged correctly as active throughout.
- **CLV and salary excluded from model features.** CLV is reserved for the downstream segmentation value axis — it is what we are trying to protect, not a predictor. Salary has near-zero correlation with churn in early checks and would make the model non-deployable in environments where salary data is unavailable.

3. Churn Definition

Churn is not a formal cancellation. We defined it more broadly — and more usefully.

A member churns if they flew zero times in H2 2018 (July–December) OR formally cancelled during that period. The prediction point T = end of June 2018. All 41 features are built from data up to and including T. Nothing from H2 2018 touches the model.

Why not use cancellation as the signal? Because cancellation is a trailing indicator. Members stop flying three to six months before they cancel. By the time cancellation appears in the records, the member has already made their decision. Using flight inactivity as the churn signal gives the model a predictable six-month intervention window.

Members with no flights in the trailing 12 months before T are excluded from the analysis. They are already disengaged and are not the target of a forward-looking retention campaign.

Result: 13,191 active members in scope. 422 churners (3.2% base rate). The model predicts which of those 13,191 members will be in the 422.

4. Model Design

Algorithm: XGBoost gradient-boosted classifier with `scale_pos_weight = 30.3`. This penalizes missing a churner 30× more than flagging a healthy member, handling the 97/3 class imbalance without oversampling — which can distort probability estimates and make risk scores unreliable for prioritization.

41 features span 8 groups, all calculated as of the prediction point T = end of June 2018:

- **Recency & Frequency** — months since last flight; current inactive streak; flight counts at 3-month, 6-month, and 12-month windows
- **Momentum & Decay** — recent vs. prior 6-month trend; recency-weighted activity score (recent months weighted more heavily than older months)
- **Distance & Intensity** — average distance per trip; average flights per active month
- **Points Engagement** — redemption ratio; ever-redeemed flag; 6-month redemption flag
- **Regularity & Seasonal** — longest inactive gap; flight volatility; peak travel season share (June–August and December)
- **Demographics** — loyalty card tier; education level; marital status; gender; enrollment type (one-hot encoded)

Validation: 5-fold stratified cross-validation on the training set. Thresholds selected on out-of-fold probabilities only — no test data was seen during tuning. Final model: 5-seed ensemble refit on all labeled data, which reduces prediction variance without overfitting.

5. Results & Validation

Model performance on the held-out test set — data the model never saw during training or tuning:

Metric	Result	Interpretation
ROC AUC	0.752	Model ranks risk correctly 75% of the time
PR AUC	0.167 vs. 0.032 baseline	5.2× lift over a random contact strategy
Top 10% capture	50% of all churners	5× lift — highly concentrated targeting
Top 20% capture	61% of all churners	Majority of disengagement in 20% of members
At 0.50 threshold	57.5% churners caught	While flagging only 13% of the member base

Top predictive features by model importance:

#	Feature	Importance	Business Meaning
1	Longest inactive streak (12m)	13%	Member went dark for months — the single strongest warning sign
2	Enrollment type: Standard vs. Promo	6%	Promo-enrolled members churn at materially higher rates
3	Points accumulation trend (6m vs. prior 6m)	3.6%	Declining earn rate flags members losing interest
4	Ever redeemed points (last 12m)	3.2%	Non-redeemers disengage far more often
5	Tenure	3.1%	Newer members more volatile; longer tenure = stickier
6	Months since last flight	3.1%	Recency alone is a strong leading indicator of dropout

Key finding: CLV is almost useless for predicting churn. High-CLV members churn at similar rates to low-CLV members when their recent behavior deteriorates. Consecutive zero-flight months and declining points engagement surface disengagement six months before cancellation appears in the records — exactly what CLV cannot see.

6. Customer Segments

Every active member is mapped to one of six segments on two axes: model risk score (High / Medium / Low) and lifetime value (top 30% vs. the rest). The matrix drives action — different segments need different responses, different urgency, and different budget.

Segment	Members	Churn Rate	Avg CLV	Recommended Action
VIP at Risk	346	25.4%	\$15,275	Personal call + companion voucher within 7 days
At Risk	873	21.9%	\$4,876	Automated re-engagement email + points bonus
High-Value, Cooling	902	4.3%	\$15,287	Quarterly lounge pass offer
Cooling	2,198	4.6%	\$4,851	Quarterly points reminder email
High-Value, Healthy	2,711	0%	\$15,203	Replace discounts with surprise-and-delight
Healthy	6,161	0%	\$4,830	No intervention needed

The segmentation reveals a counterintuitive distribution: the 2,711 High-Value Healthy members contribute the most revenue and require the least intervention. The 346 VIP at-Risk members are the most urgent — high CLV, high churn probability, small enough to contact personally this week.

7. Business Recommendations

RECOMMENDATION 1 — URGENT

Protect the 346 VIP Members — Execute Within 7 Days

The 346 VIP at-risk members represent \$5.3M in lifetime value at a 25% disengagement rate. A personal call plus a companion voucher costs roughly \$150 per member — \$52K total. If it retains 30% of them, that is \$1.6M in CLV protected on a \$52K spend.

Return: 30:1. Owner: Retention team. Timeline: Execute within 7 days.

RECOMMENDATION 2 — AUTOMATE

Build an Automated Re-Engagement Pipeline for 3,100 Cooling Members

The 3,100 Cooling members drift at 4–5% churn rates. Without intervention, roughly 145 disengage per cycle. A quarterly automated email — lounge passes for High-Value Cooling members; points reminders for standard Cooling — costs less than \$10K to set up and runs with zero ongoing headcount. This is the high-leverage, low-cost retention layer.

Cost: <\$10K setup. Headcount: zero. Owner: Marketing automation. Timeline: Next quarter.

RECOMMENDATION 3 — STOP

Stop Discounting Healthy High-Value Members

The 2,711 High-Value Healthy members carry 0% predicted churn risk. They are loyal and flying. Yet they still receive promotional discount emails — eroding margin without any retention benefit. The model predicts they will not churn regardless of whether they receive those offers.

Replace discount emails with quarterly surprise-and-delight touchpoints: priority boarding notifications, birthday recognition, exclusive status acknowledgment. These reinforce loyalty without training high-value members to expect price cuts.

Benefit: Direct margin improvement. Owner: CMO. Timeline: Next campaign cycle.

*This system identifies who to retain, how urgently, and what action delivers the best return.
It replaces reactive outreach with a systematic, repeatable retention operation.*